

3.1 Problem

observed Tx topology

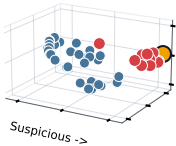


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focal 1-hop suspicious

3.2 Behavior View

$\mathcal{X}^{bhv}$ feature space

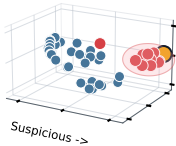


no edges yet

suspicious-side PC

3.3 Graph Recover

behavioral kNN links



top-10

recovered neighbors

3.4 GNN Encode

build $\mathcal{G}_{bhv}$ branch

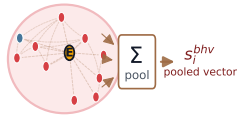


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suspicious neighbors

3.5 Subgraph Pooling

repaired $\mathcal{N}(v)$

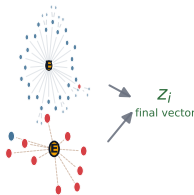


90%

pooled suspicious subgraph

3.6 Contrastive Learning

BYOL align + concat



$$z_i = [s_i^{tx} \| s_i^{bhv}]$$

final ranking representation

● ego ● suspicious ● benign — transaction edge - - behavioral kNN edge